

Qatar-2b

R-1923

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140328 · created 2026-08-06T00:58:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456744.77078 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.065
Transit depth (Rp/Rs)^2	1.27 +/- 1.47 [%]
Orbital Inclination (inc)	84.04 +/- 1.73
Airmass coefficient 1 (a1)	0.75 +/- 0.48
Airmass coefficient 2 (a2)	0.23 +/- 0.36
Scatter in the residuals of the lightcurve fit is	72.09 %
Best Comparison Star	#3 - [211, 123]
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.056 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.113 $\pm$ 0.065 vs 0.16327 (deviation 0.77 σ)
Duration fit vs published	0.056 d vs 0.0767 d (deviation 1.59 σ)
Mid-time O-C	-0.0 min
Depth SNR	1.7
Residual scatter	72.09 %

Light curve

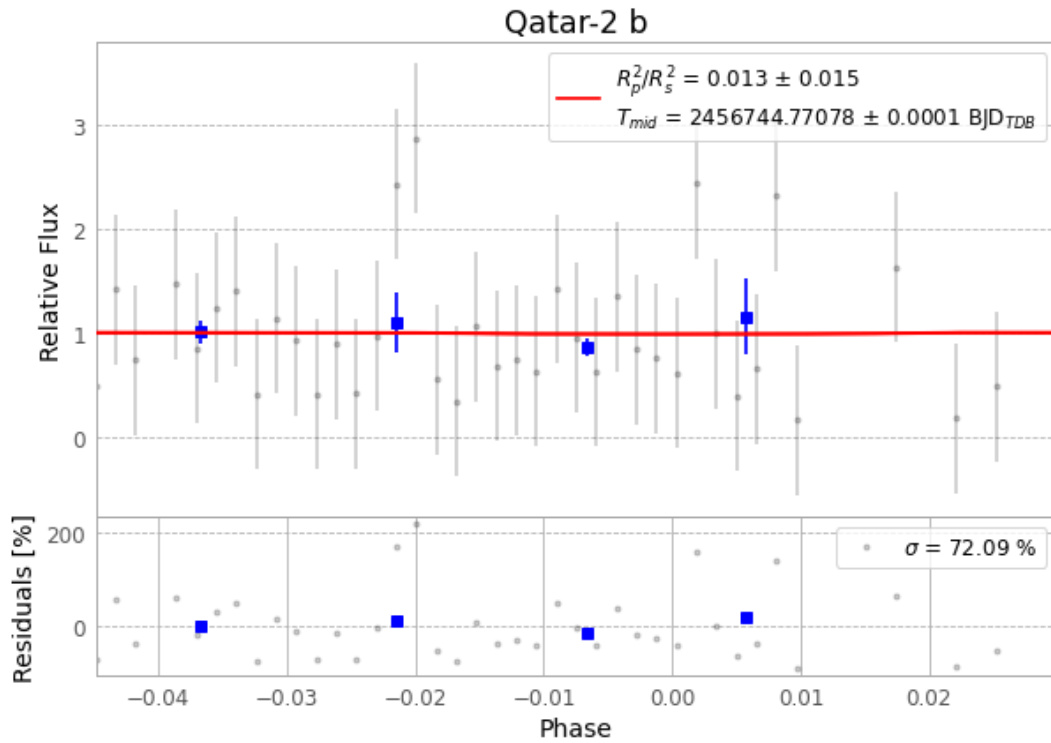


Figure 1 — FinalLightCurve_Qatar-2 b_2014-03-27.png (SHA-256 511aec967e42ed5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [146, 451], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 cc870a910da679ec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), Qatar-2140328045412.FITS → Qatar-2140328082713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140328.log (cf591630ff55...), FinalLightCurve_Qatar-2 b_2014-03-27.png (511aec967e4...), FinalLightCurve_Qatar-2 b_2014-03-27.csv (45f40ec775dc...)

Compute resources

Wall time 112.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:58Z.